

12	C	Assume that temperature is constant, then the product of pressure and volume, pV is constant. Further assume that the cross-sectional area of the tube is constant, then pl is constant. When at angle θ, $p = P_{\text{atm}} + \rho_{\text{Hg}}gl_{\text{Hg}} \cos \theta$ When the tube is upright (0°), $p = P_{\text{atm}} + \rho_{\text{Hg}}gl_{\text{Hg}}$, where Hg is mercury, pressure is maximum, l is minimum. When the tube is horizontal (90°), and $p = P_{\text{atm}}$ When the tube is inverted (180°), $p = P_{\text{atm}} - \rho_{\text{Hg}}gl_{\text{Hg}}$, pressure is minimum, l is maximum. As angle varies from 0° to 360°, l varies sinusoidally, following an inverted cosine function.
13	C	(nB)